APEX GLOBAL BANK

SR 11-7 MODEL DEVELOPMENT DOCUMENT

Stressed Value-at-Risk

Model ID	APEX-MDL-0002
Version	1.0
Owner	Dr. Priya Nair (CRO)
Validator	Dr. Samuel Achebe (Model Validation Officer)
Status	In Validation
Last Review	2026-03-01
Next Review	2027-03-01

Regulatory Framework: Basel 2.5 MRA (BCBS Jul 2009), SR 11-7

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1 Model Overview

1.1 Purpose

Stressed VaR (SVaR) supplements the standard VaR model by re-running it on a historical window calibrated to a period of significant financial stress. SVaR addresses the criticism that VaR calibrated to a calm period dramatically understates tail risk.

1.2 Business Use

- **Regulatory capital (IMA banks):** Capital $= \max(\mathrm{VaR}_t, 3 \times \overline{\mathrm{VaR}}_{60}) + \max(\mathrm{SVaR}_t, 3 \times \overline{\mathrm{SVaR}}_{60})$, per Basel 2.5.
- **Stress testing:** SVaR provides the stressed losses component in DFAST severely adverse.
- **Model governance:** SVaR independently challenges standard VaR — divergence triggers review.

1.3 Scope

Same instrument universe as APEX-MDL-0001. The stressed window uses 2007-07-01 to 2009-06-30 (GFC) as the primary calibration period.

2 Theoretical Basis

2.1 Stressed Period Selection

The BCBS requires the stressed period to capture "a period of significant financial stress relevant to the firm's portfolio." Apex uses the 2007–2009 GFC period:

- Equity markets: S&P 500 peak-to-trough -57%
- Credit spreads: IG CDS spread $+450$ bps
- FX volatility: EUR/USD realised vol $+18$ p.a.
- Rates: US10Y 2-year range > 200 bps

2.2 Methodology

SVaR uses the same Monte Carlo Cholesky method as APEX-MDL-0001, with shocked return series and stressed covariance matrix calibrated to 2007–2009 realised correlations and volatilities:

$$\mathrm{SVaR} = -\mathrm{Percentile}\big(\mathrm{P\&L} \mid \Sigma_{\mathrm{stressed}},\ \mu_{\mathrm{stressed}},\ 2000 \text{ paths},\ 99\%\big)$$

2.3 Basel 2.5 Capital Formula

$$\begin{aligned}\mathrm{IMA_capital} = \max\big(\mathrm{VaR}_t,\ k_1\ \overline{\mathrm{VaR}}_{60}\big) \times \sqrt{10} \\ + \max\big(\mathrm{SVaR}_t,\ k_2\ \overline{\mathrm{SVaR}}_{60}\big) \times \sqrt{10}\end{aligned}$$

where k_1 and k_2 are the respective backtesting multipliers (minimum 3.0 each).

3 Mathematical Specification

Parameter	GFC Stressed Value
Equity return shock (daily σ)	3.2% (vs 1.2% normal)
FX return shock	0.9%
Rates return shock (bps/day)	8.5 bps
Equity-equity correlation	0.87
Equity-credit correlation	−0.72
SVaR / VaR ratio (typical)	2.5–4.0×

4 Implementation

Code location: `infrastructure/risk/var_calculator.py` — uses `correlation_regime.STRESS` matrix with GFC-calibrated volatilities.

Stressed period re-identification: Must occur at minimum annually. Re-identification tests 3 alternative 250-day windows; selects the window producing the highest portfolio SVaR. Approved by CRO and Model Validation.

Runtime: Same as standard VaR MC (∼50 ms). Typically run daily post-close.

5 Validation

SVaR does not have a backtesting requirement under Basel 2.5 (only VaR is backfitted). Validator independently verified that GFC window reproduces reported SVaR within ±5%.

SVaR/VaR ratio monitoring: Internal guideline: SVaR/VaR ratio should be $\geq 2\times$ in normal conditions. Ratio below 1.5× triggers re-identification review.

6 Model Limitations

1. **Static stressed period:** GFC 2007–2009 may not be representative for an equity-heavy book in a tech-sector stress; COVID-2020 window produces higher equity SVaR.
2. **Annual re-identification frequency:** Period between re-identifications may allow stale period to persist through an emerging stress.
3. **No account of correlation breakdown:** SVaR correlation matrix is static; in reality, correlations during stress are dynamic.

7 Open Findings

ID	Severity	Description	Status
SVAR-F1	**Major**	Stressed period identification not re-certified since 2024-01; COVID-2020 window may produce higher SVaR than GFC for current portfolio composition.	Open